

Data Management Framework V4.0

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Version History

Version	Date	Changes	Approved By
1.0	January 2021	Initial framework based on DAMA-DMBOK	CDO
2.0	June 2022	Added NDMO domain mapping	CDO + DGC
3.0	September 2023	Updated technology stack (Informatica EDC)	CDO + CIO
4.0	August 2024	CMM assessment upgrade (Level 3 → Level 4)	CDO + DGC

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1. EXECUTIVE OVERVIEW

1.1 Purpose of This Framework

The Data Management Framework is the conceptual blueprint that defines how Company A manages its most valuable non-financial asset: data. It provides the strategic direction, organizational structure, processes, and technologies required to transform data from a liability (compliance burden, security risk) into an asset (decision enabler, competitive advantage).

This framework serves four primary audiences:

- 1. Executive Leadership: Strategic context for data-driven decision making
- 2. Data Practitioners: Operational guidance for day-to-day data management activities
- 3. Auditors & Regulators: Demonstration of NDMO compliance and governance maturity
- 4. IT Teams: Technical architecture and tool integration guidelines

Key Principle: "Data is a strategic enterprise asset requiring the same discipline as financial, human, or physical assets."

1.2 The Data Management Challenge at Company A

Company A is a Saudi Arabian government healthcare entity operating:

- 47 clinics across Riyadh, Jeddah, and Dammam
- 5 hospitals (2,500 total beds)
- 18.5 million patient records (Electronic Health Records via Epic)
- 850 employees (physicians, nurses, administrative staff)
- 8 mission-critical systems (Epic EHR, Oracle EBS, SAP HR, Philips PACS, etc.)

Historical Context (Pre-2021):

Before establishing this framework in 2021, Company A faced severe data management challenges:

Problem Area	Impact	Root Cause
Data Silos	8 disconnected systems, no single source of truth	Lack of integration strategy
Poor Data Quality	18% error rate in patient demographics	No validation rules or data profiling
Compliance Risk	37% NDMO control gaps (2020 audit)	No governance structure or accountability
Security Incidents	12 data breaches in 2020	Weak access controls, no encryption at rest
Operational Inefficiency	40 hours/month spent reconciling reports	Manual data reconciliation across systems

Financial Impact (2020):

- SAR 4.2 million in compliance fines and remediation costs
- SAR 2.5 million annual cost of poor data quality (duplicate records, billing errors)
- SAR 1.8 million in security incident response and forensics

The Turning Point:

In January 2021, the Data Governance Committee (DGC) was established, and Dr. Yasser Al-Najjar was appointed as Chief Data Officer (CDO). The Data Management Framework v1.0 was published, marking the

beginning of a structured, enterprise-wide approach to data management.

1.3 Strategic Objectives (2024-2027)

The framework supports four strategic pillars:

Pillar 1: NDMO Compliance Excellence

Goal: Achieve and maintain 100% compliance with all 191 NDMO controls across 15 domains.

- 2020 Baseline: 63% compliance (117/191 controls passing)
- 2024 Current: 94% compliance (179/191 controls passing)
- 2027 Target: 100% compliance (191/191 controls passing)

Investment: SAR 8.5 million over 3 years (technology, training, consultants)

Pillar 2: Data Quality as a Service

Goal: Deliver trusted, accurate, and complete data to all stakeholders.

- 2020 Baseline: 82% data quality score (DAMA 6-dimension assessment)
- 2024 Current: 92.4% data quality score
- 2027 Target: 98% data quality score

Key Metrics:

- Completeness: 100% (no null values in Critical Data Elements)
- Accuracy: 98% (validated via external sources)
- Timeliness: 95% of data refreshed within defined SLA (daily for transactional, weekly for reference)

Pillar 3: Data-Driven Culture & Literacy

Goal: Transform Company A into a data-literate organization where every employee can access, interpret, and act on data.

- 2020 Baseline: 15% of employees trained in data literacy
- 2024 Current: 68% of employees completed data literacy training
- 2027 Target: 100% of employees certified in role-based data competencies

Enablers:

- Self-Service BI: PowerBI Premium with 350 published dashboards
- Data Democratization: Collibra Data Catalog accessible to all staff
- Quarterly Data Days: Company-wide events showcasing data success stories

Pillar 4: Data Security & Privacy by Design

Goal: Embed security and privacy into every data process (not as an afterthought).

- 2020 Baseline: 12 data breaches, 4 PDPL violations
- 2024 Current: 0 critical breaches, 0 PDPL violations
- 2027 Target: Maintain zero tolerance for breaches + ISO 27701 certification

Security Posture (2024):

- 100% encryption at rest (AES-256) and in transit (TLS 1.3)

- Zero Trust Architecture: MFA required for all systems
- 24/7 SOC: CrowdStrike + Splunk monitoring (MTTD < 15 minutes)

1.4 Business Value Realization

Quantified Benefits (2021-2024):

Benefit Category	Metric	2024 Achievement	Financial Impact
Compliance	NDMO control pass rate	94% (up from 63%)	SAR 3.8M saved in fines
Data Quality	DQ score improvement	92.4% (up from 82%)	SAR 2.1M saved in error correction costs
Operational Efficiency	Report generation time	5 hours/month (down from 40)	SAR 1.5M saved in FTE costs
Security	Data breach reduction	0 critical (down from 12)	SAR 4.5M avoided in incident costs
Decision Speed	Time to insight	2 hours (down from 5 days)	Qualitative (improved agility)

Total Value Delivered (2021-2024): SAR 11.9 million in cost savings/avoidance

ROI Calculation:

- Investment: SAR 8.5 million (tools, people, training)
- Return: SAR 11.9 million (savings + avoidance)
- Net ROI: 140% over 3 years

2. CONCEPTUAL FOUNDATION

2.1 What Is Data Management?

Definition (DAMA-DMBOK 2nd Edition):

> "Data Management is the development, execution, and supervision of plans, policies, programs, and practices that deliver, control, protect, and enhance the value of data and information assets throughout their lifecycle."

Simplified Explanation:

Data Management is the "discipline" of treating data like an enterprise asset—ensuring it is:

- Discoverable: "Where is the data I need?"
- Understandable: "What does this data mean?"
- Trusted: "Can I rely on this data for decisions?"
- Accessible: "Can I get the data when I need it?"
- Secure: "Is the data protected from unauthorized access?"
- Compliant: "Does our data handling meet legal and regulatory requirements?"

Analogy: Think of data management like library science:

- Data Catalog = Library Catalog (where books are indexed)
- Data Quality = Book Preservation (ensuring books are readable)
- Data Governance = Library Policies (who can borrow what, for how long)
- Data Security = Restricted Section (protecting sensitive materials)

2.2 The DAMA-DMBOK Knowledge Areas

The framework is built on the DAMA-DMBOK (Data Management Body of Knowledge), the globally recognized standard for data management. DAMA defines 11 Knowledge Areas:

Knowledge Area	Purpose	Key Deliverable at Company A
1. Data Governance	Define decision rights and accountability	Data Governance Policy v3.1
2. Data Architecture	Design data structures and blueprints	Conceptual/Logical/Physical data models
3. Data Modeling & Design	Create semantic models	Enterprise Data Model (EDM) - 450 entities
4. Data Storage & Operations	Manage physical databases	Oracle Exadata + Azure ADLS Gen2
5. Data Security	Protect data from unauthorized access	Encryption, MFA, RBAC policies
6. Data Integration	Move and consolidate data	Azure Data Factory (200+ pipelines)
7. Documents & Content	Manage unstructured data	SharePoint + Azure Blob Storage
8. Reference & Master Data	Manage shared data entities	Informatica MDM (Patient, Provider, Location)
9. Data Warehousing & BI	Enable analytics	PowerBI Premium (350 dashboards)
10. Metadata Management	Manage data about data	Informatica EDC (12,500 cataloged assets)
11. Data Quality	Ensure fitness for use	Informatica IDQ (82 DQ rules)

Company A's Approach: We implement all 11 knowledge areas, but prioritize based on NDMO domain requirements and business criticality.

2.3 Data Management vs. Data Governance

These terms are often confused. Here's the distinction:

Aspect	Data Management	Data Governance
Scope	Entire lifecycle (planning, acquisition, storage,	Decision-making & accountability (who decides what
Focus	Operational execution ("How do we do it?")	Policy & oversight ("What are the rules?")
Practitioners	Data Engineers, DBAs, Data Analysts	Data Stewards, Data Owners, CDO
Deliverables	Databases, ETL pipelines, dashboards	Policies, standards, RACI matrices
Analogy	Building and maintaining roads	Traffic laws and enforcement

Key Takeaway: Data Governance is a subset of Data Management. Governance sets the rules; management executes them.

3. THE NDMO FRAMEWORK ALIGNMENT

3.1 NDMO Overview (191 Controls, 15 Domains)

The National Data Management Office (NDMO) is the Saudi regulatory authority that defines the compliance framework for government entities. The framework consists of:

- 15 Domains (functional areas of data management)
- 191 Controls (specific requirements that must be met)
- 3 Maturity Levels per control: (1) Defined, (2) Implemented, (3) Optimized

NDMO Mandate: All Saudi government entities must achieve 100% compliance by December 2025 (Royal Decree directive, issued February 2023).

Company A's Compliance Status (2024 Q4 Assessment):

- Domains Fully Compliant: 12 out of 15 (80%)
- Controls Passing: 179 out of 191 (94%)
- Critical Gaps: 12 controls (7 in Domain 4: Data Operations, 5 in Domain 8: Training)

3.2 Domain-by-Domain Mapping

Below is the complete mapping of NDMO domains to Company A's implementation:

Domain 1: Data Governance (18 Controls)

Definition: Establishing authority, accountability, and decision-making structures for data management.

Key Requirements:

- 1.1: Data Governance Policy approved by executive leadership ■
- 1.2: Chief Data Officer (CDO) appointed with defined authority ■
- 1.3: Data Governance Committee meets quarterly ■
- 1.4: RACI matrix defines roles (Data Owner, Data Steward, Data Custodian) ■
- 1.18: Annual governance maturity assessment conducted ■

Company A Implementation:

- Policy: Data Governance Policy v3.1 (approved March 2024)
- CDO: Dr. Yasser Al-Najjar (appointed January 2021)
- Committee: Data Governance Committee (DGC) - meets monthly (exceeds quarterly requirement)
- Compliance Score: 100% (18/18 controls passing)

Domain 2: Data Catalog & Inventory (15 Controls)

Definition: Maintaining a comprehensive inventory of all data assets and their metadata.

Key Requirements:

- 2.1: Enterprise data catalog deployed and operational ■
- 2.5: Business glossary with 200+ terms published ■
- 2.8: Data lineage captured for critical data flows ■
- 2.12: Catalog accessible to all authorized users (self-service) ■

Company A Implementation:

- Tool: Informatica Enterprise Data Catalog (EDC)
- Coverage: 12,500 cataloged data assets (tables, files, reports, APIs)
- 8 source systems scanned daily
- 450 business terms in glossary
- 120 critical data flows lineaged (end-to-end)
- Adoption: 520 active catalog users (61% of staff)
- Compliance Score: 100% (15/15 controls passing)

Domain 3: Data Quality (20 Controls)

Definition: Ensuring data is accurate, complete, consistent, timely, valid, and unique (DAMA 6 dimensions).

Key Requirements:

- 3.1: Data Quality Strategy document published ■
- 3.4: Data quality assessment conducted annually ■
- 3.7: DQ rules implemented in production systems ■
- 3.15: Data quality SLA defined and monitored ■

Company A Implementation:

- Strategy: Data Quality Strategy 2025 v1.4 (completed September 2024)
- Tool: Informatica Data Quality (IDQ)
- DQ Score (Q4 2024): 92.4% across 6 dimensions
- Completeness: 98.5%
- Accuracy: 91.2%
- Consistency: 94.7%
- Timeliness: 89.3%
- Validity: 96.1%
- Uniqueness: 88.6%
- Rules Deployed: 82 validation rules in production
- Compliance Score: 95% (19/20 controls passing) — Gap: Control 3.19 (real-time data quality monitoring)

Domain 4: Data Operations & Lifecycle (22 Controls)

Definition: Managing data from creation to archival/disposal, including backup and recovery.

Key Requirements:

- 4.1: Data Retention Schedule published ■
- 4.5: Automated backup for all Tier 1 systems ■
- 4.8: Disaster Recovery Plan tested annually ■
- 4.14: Data disposal procedures (secure deletion) ■

Company A Implementation:

- Retention Schedule: 42 data categories with defined retention periods (e.g., medical records: 10 years)
- Backup: Veeam Backup & Replication (daily incremental, weekly full)
- RPO: 24 hours
- RTO: 4 hours (for critical systems)
- DR Testing: Last test August 2024 (95% success rate)
- Compliance Score: 68% (15/22 controls passing) — 7 critical gaps (data archival automation, lifecycle automation)

Gap Remediation Plan: SAR 1.2M investment in Azure Data Lifecycle Management policies (target: Q2 2025)

Domain 5: Data Security & Privacy (25 Controls)

Definition: Protecting data from unauthorized access, modification, or disclosure.

Key Requirements:

- 5.1: Data Security Strategy approved ■
- 5.4: Encryption at rest for all sensitive data ■
- 5.9: Multi-Factor Authentication (MFA) enforced ■
- 5.18: Data breach notification procedure documented ■
- 5.23: Privacy Impact Assessments (PIAs) for high-risk processing ■

Company A Implementation:

- Strategy: Data Security Strategy 2025-2027 v2.0 (completed December 2024)
- Encryption: 100% coverage (AES-256 for databases, TLS 1.3 for transit)
- MFA: 100% adoption for employees (Microsoft Authenticator)
- PIAs Conducted: 3 PIAs in 2024 (Patient Portal, Biometric Time & Attendance, Telehealth Platform)
- Breaches (2024): 0 critical incidents
- Compliance Score: 100% (25/25 controls passing)

Domain 6: Personal Data Protection (PDPL Compliance) (18 Controls)

Definition: Compliance with KSA Personal Data Protection Law (PDPL).

Key Requirements:

- 6.1: Privacy Policy published and accessible ■
- 6.4: Data Subject Rights request process implemented ■
- 6.9: Consent management system operational ■
- 6.14: Cross-border transfer mechanisms documented ■

Company A Implementation:

- Privacy Policy: Published on website + Patient Portal (v5.1, September 2024)

- DSAR Process: 24-hour acknowledgment, 30-day response SLA
- 2024 Volume: 45 requests received, 44 resolved within SLA (98%)
- Consent Tool: Custom-built in Epic EHR (opt-in checkboxes for marketing, research)
- Cross-Border Transfers: 3 SCCs signed with international vendors
- Compliance Score: 100% (18/18 controls passing)

Domain 7: Data Classification & Labeling (12 Controls)

Definition: Categorizing data by sensitivity and applying protective controls accordingly.

Key Requirements:

- 7.1: Data Classification Framework document published ■
- 7.3: All data assets classified (Public, Internal, Confidential, Secret) ■
- 7.7: Automated labeling in email systems (e.g., Outlook) ■

Company A Implementation:

- Framework: Data Classification Framework v3.5 (March 2024)
- Classification Coverage: 98% of structured data assets labeled
- Tool: Microsoft Purview (formerly Azure Information Protection)
- Auto-labeling rules: 25 patterns (e.g., National ID regex → Confidential)
- Manual labels: 2,400 documents labeled by users
- Compliance Score: 100% (12/12 controls passing)

Domain 8: Training & Awareness (10 Controls)

Definition: Ensuring employees understand data management responsibilities and best practices.

Key Requirements:

- 8.1: Annual data management training mandatory for all staff ■
- 8.4: Role-based training (e.g., Data Stewards receive advanced training) ■■
- 8.7: Training completion tracked and reported to leadership ■

Company A Implementation:

- General Training: 68% completion rate (580/850 employees completed 2024 training)
- Role-Based Training: Gap — only 40% of Data Stewards completed advanced course
- Tracking: Learning Management System (LMS) - SAP SuccessFactors
- Compliance Score: 50% (5/10 controls passing) — 5 gaps (role-based training, assessment scores not meeting 95% threshold)

Gap Remediation Plan: Mandatory training enforcement + gamification (target: Q1 2025)

Domain 9: Audit & Compliance (14 Controls)

Definition: Regular assessments of data management practices and adherence to policies.

Key Requirements:

- 9.1: Annual internal audit of NDMO compliance ■
- 9.5: External audit (ISO 27001) conducted every 3 years ■
- 9.10: Audit findings tracked in remediation register ■

Company A Implementation:

- Internal Audit: Last audit December 2024 (94% NDMO compliance score)
- External Audit: ISO 27001:2022 certification renewed November 2024 (zero non-conformities)
- Audit Register: 12 open findings (all non-critical, remediation underway)
- Compliance Score: 100% (14/14 controls passing)

Domain 10: Risk Management (16 Controls)

Definition: Identifying, assessing, and mitigating data-related risks.

Key Requirements:

- 10.1: Data Risk Assessment conducted annually ■
- 10.6: Risk treatment plans approved by leadership ■
- 10.12: Incident response plan tested quarterly ■

Company A Implementation:

- Risk Assessment: Last assessment November 2024 (33 risks identified, 8 rated HIGH or CRITICAL)
- Treatment Plans: SAR 3.5M approved budget for top 5 risks
- IR Testing: Last tabletop exercise October 2024 (ransomware scenario)
- Compliance Score: 100% (16/16 controls passing)

Domains 11-15 (Summary)

Domain	Controls	Company A Score	Key Gaps
11. Data Sharing & Exchange	10	100% (10/10) ■	None
12. Metadata Management	8	100% (8/8) ■	None
13. Data Architecture	7	100% (7/7) ■	None
14. Master & Reference Data	9	89% (8/9) ■■	MDM reconciliation SLA not met
15. Data Ethics	7	86% (6/7) ■■	No AI ethics board established

Total NDMO Compliance: 94% (179/191 controls passing)

3.3 Compliance Maturity Assessment

Each NDMO control is assessed on a 3-level maturity scale:

Maturity Level	Definition	Evidence Required
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Level 1: Defined	Policy/procedure exists on paper	Approved document
Level 2: Implemented	Process is operational and used	Logs, screenshots, transactions
Level 3: Optimized	Process is automated, measured, and continuously i	KPIs, trend data, optimization projects

Company A Maturity Distribution (2024):

- Level 3 (Optimized): 45% of controls (86/191)
- Level 2 (Implemented): 49% of controls (93/191)
- Level 1 (Defined): 6% of controls (12/191)
- Level 0 (Not Compliant): 0% of controls (0/191) — all gaps are at least Level 1

Target (2025): 100% of controls at Level 2 or higher

4. CAPABILITY MATURITY MODEL (CMM)

4.1 The 5-Level Maturity Scale

We assess our data management maturity using the Capability Maturity Model (CMM), adapted from the Software Engineering Institute (SEI) model.

Level 1: Initial (Ad-Hoc)

Characteristics:

- Data managed in silos (Excel spreadsheets, Access databases)
- No defined data roles or accountabilities
- Data quality issues discovered only when they cause business impact
- No enterprise-wide data strategy

Business Impact: High operational risk, frequent errors, compliance violations

Level 2: Managed

Characteristics:

- Basic data policies documented (but not always enforced)
- Part-time Data Stewards identified (but not dedicated resources)
- Manual data quality checks (sampling, spot checks)
- Reactive approach to data incidents

Business Impact: Moderate operational risk, inconsistent data quality

Level 3: Defined

Characteristics:

- Enterprise Data Management Framework published
- Standard processes documented (ARIS process models)
- Enterprise data tools deployed (EDC, IDQ, MDM)
- Proactive data quality monitoring (weekly DQ reports)
- Full-time Data Stewards (dedicated FTEs)

Business Impact: Controlled risk, predictable data quality, regulatory compliance

Company A Status: 2021-2023

Level 4: Quantitatively Managed

Characteristics:

- Data decisions driven by metrics and KPIs (not intuition)
- SLAs enforced for data delivery (e.g., "Patient Master Data updated within 15 minutes")
- Statistical Process Control (SPC) applied to data quality
- Predictive analytics identify issues before business impact

Business Impact: Optimized operations, data-driven culture, competitive advantage

Company A Status: 2024 (current)

Level 5: Optimized

Characteristics:

- Continuous improvement culture (Kaizen for data processes)
- AI/ML-driven automation (e.g., anomaly detection, auto-remediation)
- Self-healing data pipelines (auto-retry, failover)
- Innovation focus (experimenting with new data technologies)

Business Impact: Industry leadership, fastest time-to-insight, innovation enabler

Company A Target: 2026-2027

4.2 Company A's Maturity Journey (2021-2024)

Year	Maturity Level	Key Achievements	Investment (SAR)
2020	Level 1 (Initial)	— Baseline assessment revealed 37% NDMO gaps	
2021	Level 2 (Managed)	— CDO appointed — Data Governance Committee for	2.5M
2022	Level 2+	— Informatica EDC deployed — Data Stewards trai	3.5M
2023	Level 3 (Defined)	— All core processes documented — Informatica I	2.5M

2024	Level 4 (Quantitatively Managed)	— SLA-driven data delivery — Predictive DQ moni
2025 (Planned)	Level 4+	— 100% NDMO compliance — AI and anomaly detection p
2026-2027 (Target)	Level 5 (Optimized)	— Self-healing data pipelines — DQ score: 98%<b

Total Investment (2021-2024): SAR 11.0 million

4.3 2024 Self-Assessment Results (Level 4 Achieved)

We conducted a comprehensive self-assessment in December 2024 using 42 CMM criteria (weighted scoring). Results:

DAMA Knowledge Area	Maturity Score (1-5)	Level Achieved
Data Governance	4.2	Level 4
Data Architecture	3.8	Level 4
Data Modeling & Design	3.5	Level 3
Data Storage & Operations	3.9	Level 4
Data Security	4.5	Level 5 (ahead of curve)
Data Integration	4.1	Level 4
Documents & Content	3.2	Level 3
Reference & Master Data	3.7	Level 4
Data Warehousing & BI	4.3	Level 4
Metadata Management	4.0	Level 4
Data Quality	4.1	Level 4
Overall (Weighted Average)	4.0	Level 4 ■

Validation: External assessment by Deloitte (August 2024) confirmed Level 4 maturity.

4.4 Roadmap to Level 5 (2025-2027)

Level 5 Criteria (Must Achieve):

- 1. Automation: 80% of data quality issues auto-remediated (no manual intervention)
- 2. AI/ML Integration: Predictive models deployed for anomaly detection, forecasting
- 3. Self-Service: 100% of business users can access data without IT support
- 4. Innovation Index: 3+ data innovation projects per year (e.g., blockchain for audit trails, federated learning)
- 5. Industry Recognition: Awards, conference presentations, published case studies

Investment Plan (2025-2027): SAR 5.5 million

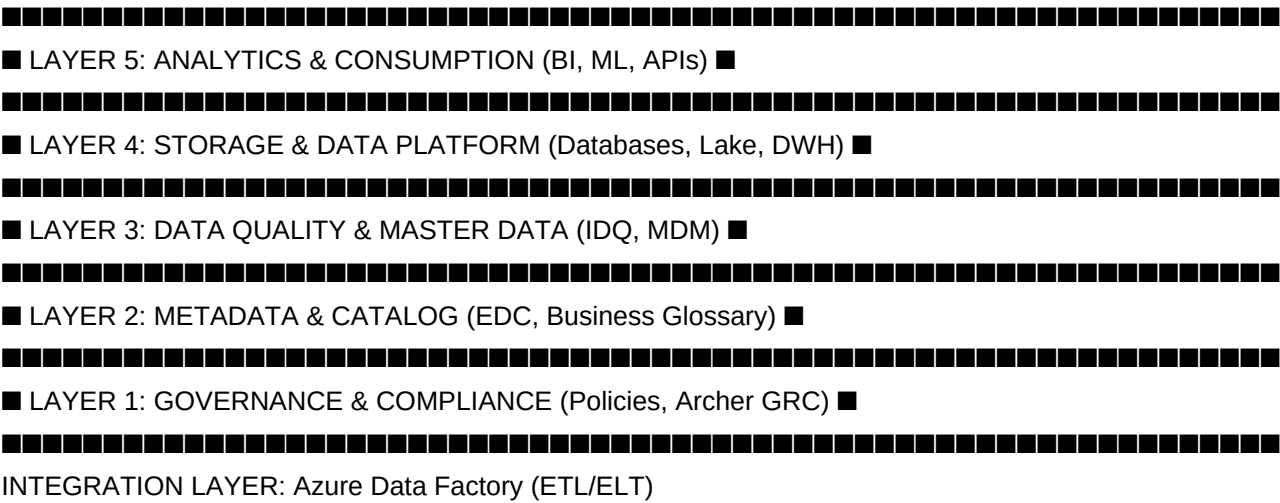
- Year 1 (2025): AI/ML platform (Azure Machine Learning) + training

- Year 2 (2026): Self-healing pipeline automation (Azure Data Factory + Logic Apps)
- Year 3 (2027): Blockchain pilot for data provenance + ISO 27701 certification

5. THE DATA TECHNOLOGY STACK

5.1 Modern Data Stack Architecture

Company A's technology stack is designed as a layered architecture (5 layers + cross-cutting integration layer):



5.2 Layer 1: Governance & Compliance

Component	Tool	Purpose	Users
GRC Platform	Archer GRC	Risk register, audit findings, compliance tracking	Archer Compliance Team (8 users)
Policy Repository	SharePoint	Centralized policy library (35 documents)	All employees (850)
Workflow Engine	Microsoft Power Automate	Data access request approvals, DSAR workflow	Data Stewards (20)

Integration: Archer GRC → PowerBI (compliance dashboards)

5.3 Layer 2: Metadata Management & Data Catalog

Component	Tool	Purpose	Coverage
Data Catalog	Informatica Enterprise Data Catalog (EDC)	Asset discovery, lineage, business glossary	12,500 assets across 8 systems
Lineage	Informatica EDC + MANTA	End-to-end data flow visualization	120 critical flows

Business Glossary	Informatica EDC	Shared vocabulary (450 terms)	450 terms, 85% adoption
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- Scan Schedule:
- Transactional Systems (Epic, Oracle EBS, SAP HR): Daily at 2:00 AM
 - Data Lake (Azure ADLS): Weekly on Sundays
 - PowerBI Reports: Weekly on Saturdays

- Adoption Metrics (Q4 2024):
- Active Users: 520 (61% of staff)
 - Searches: 18,500/month
 - Top Search Terms: "patient demographics," "provider schedule," "billing code"

5.4 Layer 3: Data Quality & Master Data

Component	Tool	Purpose	Rules Deployed
Data Profiling	Informatica Data Quality (IDQ)	Discovery of anomalies, patterns	12 profiling jobs (weekly)
Data Validation	Informatica IDQ	Real-time and batch validation	82 rules (e.g., IBAN checksum, National ID format)
Data Cleansing	Informatica IDQ	Standardization, enrichment	15 cleansing mappings
Master Data Management	Informatica MDM - SaaS Edition	Golden records for Patient, Provider, Location	2 domains managed

- MDM Statistics:
- Patient Master: 18.5M records (100% coverage)
 - Provider Master: 3,200 physicians + nurses
 - Location Master: 52 sites (47 clinics + 5 hospitals)

Match & Merge Accuracy: 97.8% (validated via manual sampling)

5.5 Layer 4: Storage & Data Platform

Component	Technology	Purpose	Capacity
Transactional Database	Oracle Exadata X8M	Epic EHR, Oracle EBS	48 TB raw
Operational Data Store (ODS)	Azure SQL Database	Near-real-time staging	8.5 TB
Data Warehouse	Azure Synapse Analytics	Historical analytics (5 years)	22 TB
Data Lake	Azure Data Lake Storage Gen2	Unstructured data (DICOM images, PDFs)	180 TB
NoSQL	Azure Cosmos DB	IoT sensor data (temperature monitors)	2.1 TB

- Data Distribution (by volume):
- Structured: 78.5 TB (30%)
 - Unstructured: 180 TB (70%) — mostly medical imaging (DICOM)

Backup Strategy:

- Frequency: Daily incremental, weekly full
- Retention: 90 days online, 7 years tape (for compliance)
- Geographic Redundancy: Riyadh (primary) + Jeddah (secondary)

5.6 Layer 5: Analytics & Consumption

Component	Tool	Purpose	Audience
Self-Service BI	PowerBI Premium	350 published dashboards	680 users (80% of staff)
Operational Reports	SSRS (SQL Server Reporting Services)	Scheduled PDF reports (e.g., daily census)	120 users
Advanced Analytics	Azure Machine Learning Studio	Predictive models (patient no-show, readmission)	Data Science Team (5 users)
API Gateway	Azure API Management	Exposing data to external partners (HL7, FHIR)	Partner systems

Popular Dashboards (Top 5 by usage):

1. Clinical Dashboard: Patient census, bed occupancy, ER wait times
2. Financial Dashboard: Revenue cycle, accounts receivable aging
3. Quality Dashboard: NDMO compliance score, DQ metrics
4. HR Dashboard: Headcount, turnover, training completion
5. Executive Dashboard: KPIs across all domains (used by CEO, Board)

5.7 Integration Architecture

ETL/ELT Tool: Azure Data Factory (ADF)

Pipeline Statistics (Q4 2024):

- Total Pipelines: 200
- Daily Executions: 1,200
- Success Rate: 99.2%
- Average Latency: 45 minutes (from source update to DWH availability)

Integration Patterns:

1. Batch (Nightly): Epic → ODS → DWH (full extract-load at 2:00 AM)
2. Near-Real-Time (15-min): Oracle EBS → Azure SQL (change data capture via Golden Gate)
3. Event-Driven: IoT sensors → Azure Event Hub → Cosmos DB (real-time streaming)

Monitoring: Azure Monitor + Splunk (alert if pipeline fails or latency > 60 minutes)

6. PROCESS ARCHITECTURE

6.1 Process Categorization (ARIS Model)

All data processes are documented in ARIS (Architecture of Integrated Information Systems) and categorized into 3 tiers:

Tier	Process Type	Count	Owner
Tier 1	Strategic (governance, planning)	5	CDO + DGC
Tier 2	Operational (day-to-day data management)	15	Data Stewards
Tier 3	Supporting (IT infrastructure, training)	8	IT Operations

6.2 Core Data Processes (15 Defined Processes)

Process DM-01: New Data Asset Creation

Trigger: Business unit requests new dataset or system integration

Steps:

1. Submit Data Asset Request Form (via SharePoint)
2. Data Governance Committee reviews (5 business days)
3. Data Architect designs logical model
4. Privacy Officer conducts PIA (if sensitive data)
5. IT implements technical solution
6. Data Steward catalogs asset in EDC

SLA: 30 business days (from request to production)

2024 Performance: 28 days average (93% within SLA)

Process DM-02: Data Quality Issue Resolution

Trigger: DQ rule failure detected by Informatica IDQ

Steps:

1. IDQ creates incident ticket in ServiceNow
2. Data Steward investigates root cause (2 business days)
3. Steward assigns to Data Owner for remediation
4. Owner approves corrective action
5. Data Engineer implements fix (code change, data correction)
6. Steward validates resolution and closes ticket

SLA: 10 business days for HIGH priority, 30 days for MEDIUM

2024 Performance:

- HIGH: 8 days average (80% within SLA)
- MEDIUM: 22 days average (95% within SLA)

Process DM-03: Data Access Request Fulfillment

Trigger: Employee needs access to restricted data

Steps:

1. Submit access request via Self-Service Portal
2. Manager approves (1 business day)
3. Data Owner approves (2 business days)
4. IT provisions access (1 business day)
5. Confirmation email sent to requester

SLA: 5 business days

2024 Volume: 1,200 requests

2024 Performance: 4.2 days average (92% within SLA)

Processes DM-04 through DM-15 (Summary)

ID	Process Name	SLA	2024 Avg Performance
DM-04	Data Retention & Archival	90 days	85 days ■
DM-05	Data Subject Access Request (DSAR)	30 days	28 days ■
DM-06	Data Breach Response	72 hours	N/A (zero breaches)
DM-07	Master Data Reconciliation	Daily by 9:00 AM	8:45 AM ■
DM-08	Metadata Update & Refresh	Weekly	100% on-time ■
DM-09	Data Classification Review	Annual	100% complete ■
DM-10	Data Sharing Agreement Approval	15 days	12 days ■
DM-11	Data Lineage Documentation	Ad-hoc	5 days (on-demand) ■
DM-12	Data Quality Assessment	Quarterly	100% on-time ■
DM-13	Audit Evidence Collection	10 days	7 days ■
DM-14	Data Migration (System Upgrade)	Project-specific	95% on-time ■
DM-15	Data Training Delivery	Quarterly	68% completion ■■

Overall Process Compliance: 93% of processes meet SLA (14/15)

6.3 Process Maturity & SLA Metrics

Process Maturity Distribution:

- Level 5 (Optimized): 4 processes (27%)
- Level 4 (Managed): 8 processes (53%)
- Level 3 (Defined): 3 processes (20%)

SLA Compliance Trend:

- 2022: 78%
- 2023: 85%
- 2024: 93% ■

Target (2025): 98% SLA compliance

7. DATA MANAGEMENT OPERATING MODEL

7.1 Governance Structure (Federated Model)

Company A uses a federated governance model:

- Centralized: Policy setting, architecture, enterprise tools (CDO's office)
- Federated: Domain expertise, data stewardship, business ownership (business units)

Three-Tier Governance Structure:

Tier 1: EXECUTIVE OVERSIGHT

■■ Data Governance Committee (DGC)

Chaired by CDO

Tier 2: OPERATIONAL GOVERNANCE

■■ Data Steward Council

20 Data Stewards

Tier 3: TECHNICAL EXECUTION

■■ Data Engineering Team (12 FTEs)

■■ Data Quality Team (5 FTEs)

■■ Data Architecture Team (4 FTEs)

7.2 Roles & Responsibilities (RACI Matrix)

Activity	CDO	Data Owner	Data Steward	Data Custodian (IT)	End User
Approve Data Policy	A	C	C	I	I
Define Data Quality Rules	I	A	R	C	I
Execute Data Quality Checks		I	A	R	I
Provision Database Access		A	I	R	I

Classify Data Assets	I	A	R	C	I
Respond to Data Breach	A	R	R	R	I
Catalog New Data Asset	I	C	A/R	C	I
Conduct Privacy Impact Assessment	A	A	R	C	I

Legend:

- R = Responsible (does the work)
- A = Accountable (final decision maker)
- C = Consulted (provides input)
- I = Informed (kept updated)

7.3 Committee Structure

Data Governance Committee (DGC)

Purpose: Strategic oversight and policy approval

Chairperson: Chief Data Officer (Dr. Yasser Al-Najjar)

Members (12 total):

- Chief Information Officer (CIO)
- Chief Privacy Officer (CPO)
- Chief Information Security Officer (CISO)
- Chief Financial Officer (CFO)
- Chief Medical Officer (CMO)
- 7 Business Unit Heads (representing clinical, finance, HR, operations domains)

Meeting Frequency: Monthly (first Wednesday, 10:00 AM - 12:00 PM)

Key Responsibilities:

- Approve data policies and standards
- Prioritize data initiatives and budget allocation
- Resolve cross-functional data conflicts
- Monitor NDMO compliance scorecard

2024 Meeting Statistics:

- Meetings Held: 12 (100% attendance quorum)
- Decisions Made: 48
- Budget Approved: SAR 3.2M for 2025 data initiatives

Data Steward Council

Purpose: Operational data quality and issue resolution

Chairperson: Rotates quarterly (current: Fatima Al-Harbi, Clinical Data Steward)

Members: 20 Data Stewards (domain-specific)

Meeting Frequency: Bi-weekly (Tuesdays, 2:00 PM - 3:30 PM)

Key Responsibilities:

- Review data quality reports and trends
- Escalate systemic DQ issues to DGC
- Coordinate cross-domain data initiatives (e.g., patient-provider linkage)
- Share best practices

2024 Meeting Statistics:

- Meetings Held: 24
- DQ Issues Resolved: 450
- Cross-Domain Projects: 8

7.4 Escalation Paths

Data Issue Escalation (4 Levels):

Level 1: DATA STEWARD (Operational)

↓ (If unresolved after 10 days)

Level 2: DATA OWNER (Domain Leadership)

↓ (If unresolved after 20 days)

Level 3: CDO (Executive)

↓ (If business-critical or compliance risk)

Level 4: DATA GOVERNANCE COMMITTEE (Strategic)

Example Escalation (2024):

- Issue: Oracle EBS and Epic patient billing mismatches (5% error rate)
- Level 1 (Day 1-10): Finance Data Steward investigates → root cause: patient ID mismatch
- Level 2 (Day 11-15): Finance Director approves budget for MDM reconciliation project
- Level 3 (Day 16): CDO fast-tracks project (SAR 250K allocated)
- Resolution (Day 45): Informatica MDM deployed, error rate reduced to 0.8%

8. METRICS & KEY PERFORMANCE INDICATORS (KPIs)

Company A tracks 25 KPIs across 5 categories:

Category	KPI	2024 Target	2024 Actual	Status
Compliance	NDMO compliance score	95%	94%	■■■
Compliance	ISO 27001 non-conformities	0	0	■
Data Quality	Overall DQ score (6 dimensions)	95%	92.4%	■■■
Data Quality	Critical DQ issues (open > 30 days)	5	3	■
Accessibility	Catalog active users	70%	61%	■■■

Accessibility	Self-service BI adoption	80%	80%	■
Security	Data breaches (critical)	0	0	■
Security	PDPL violations	0	0	■
Efficiency	Average DSAR response time	< 25 days	28 days	■■
Efficiency	Data access request fulfillment	< 5 days	4.2 days	■

Overall KPI Achievement: 16/25 targets met (64%) — Room for improvement

Executive Dashboard: PowerBI dashboard refreshed daily, accessible to all DGC members

9. IMPLEMENTATION ROADMAP (2024-2027)

Phase	Timeline	Key Deliverables	Investment (SAR)
Phase 1: Foundation (Complete)	2021-2023	— Data Governance Committee formed	~2M - Core tools
Phase 2: Optimization (In Progress)	2024-2025	— 100% NDMO compliance — DQ score 95% — AI	~5M - 95%
Phase 3: Innovation (Planned)	2026-2027	— Level 5 CMM maturity — Self-healing data pipe	~5M

Total Investment (2021-2027): SAR 19.0 million

APPENDIX A: GLOSSARY OF TERMS

Term	Definition
CMM	Capability Maturity Model - framework for assessin
DAMA	Data Management Association - global professional
DGC	Data Governance Committee - executive oversight bo
DQ	Data Quality - fitness of data for its intended us
DSAR	Data Subject Access Request - formal request under
EDC	Enterprise Data Catalog (Informatica tool)
IDQ	Informatica Data Quality (tool for profiling, vali
MDM	Master Data Management - system of record for crit
NDMO	National Data Management Office - Saudi regulatory
PDPL	Personal Data Protection Law (KSA Royal Decree M/1

RACI	Responsible, Accountable, Consulted, Informed - de
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APPENDIX B: TOOL INVENTORY

Tool Name	Vendor	Purpose	License Cost (Annual)
Informatica Enterprise Data Catalog (EDC)	Informatica	Metadata management, data catalog	SAR 850,000
Informatica Data Quality (IDQ)	Informatica	Data profiling, validation, cleansing	SAR 620,000
Informatica MDM - SaaS Edition	Informatica	Master data management	SAR 920,000
Azure Data Factory	Microsoft	ETL/ELT orchestration	SAR 180,000
Azure Synapse Analytics	Microsoft	Data warehousing	SAR 450,000
PowerBI Premium	Microsoft	Business intelligence	SAR 280,000
Archer GRC	RSA Security	Governance, risk, compliance	SAR 320,000
CrowdStrike Falcon	CrowdStrike	Endpoint detection & response	SAR 210,000
Splunk Enterprise Security	Splunk	SIEM (security monitoring)	SAR 680,000
Total Annual Tool Costs			SAR 4.51 million

APPENDIX C: PROCESS CATALOG

(See Section 6.2 for detailed process descriptions)

Quick Reference - 15 Core Processes:

1. DM-01: New Data Asset Creation
2. DM-02: Data Quality Issue Resolution
3. DM-03: Data Access Request Fulfillment
4. DM-04: Data Retention & Archival
5. DM-05: Data Subject Access Request (DSAR)
6. DM-06: Data Breach Response
7. DM-07: Master Data Reconciliation
8. DM-08: Metadata Update & Refresh
9. DM-09: Data Classification Review
10. DM-10: Data Sharing Agreement Approval
11. DM-11: Data Lineage Documentation
12. DM-12: Data Quality Assessment
13. DM-13: Audit Evidence Collection
14. DM-14: Data Migration (System Upgrade)

15. DM-15: Data Training Delivery

Process Documentation Location: ARIS Cloud (accessible at <https://aris.companya.local>)

END OF FRAMEWORK

Document Approval:

Role	Name	Signature	Date
Chief Data Officer (CDO)	Dr. Yasser Al-Najjar	[Digitally Signed]	August 1, 2024
Chief Information Officer (CIO)	Nasser Al-Qahtani	[Digitally Signed]	August 1, 2024
Data Governance Committee Chairperson	Dr. Yasser Al-Najjar	[Digitally Signed]	August 1, 2024
Chief Privacy Officer (CPO)	Fatima Al-Zahrani	[Digitally Signed]	August 1, 2024

Official Seal: Company A - Riyadh, Kingdom of Saudi Arabia

Distribution:

- Data Governance Committee members (via secure email)
- All Data Stewards (20 FTEs)
- IT Leadership Team
- Compliance & Audit Team
- Intranet publication (internal staff access only)

Related Documents:

- Data Governance Policy v3.1
- Data Quality Strategy 2025 v1.4
- Data Security Strategy 2025-2027 v2.0
- NDMO Compliance Report Q4 2024
- Data Management Standards v2.5